ASSIGNMENT 4 APPENDIX: EVIDENCE & PORTFOLIO

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Section 1: Week 8 Ratio Calculations & Financial Discrepancy Audit

Independent calculation of KMG 2025 financial ratios compared against AI-supported initial output, detailing discrepancy identification and resolution.

Financial Metric / Ratio	KMG 2025 Primary Source	AI Initial Output	Human Audit & Discrepancy Resolution	Verified Figure
Net Debt / EBITDA	KMG Fin. Statements Note 6 Net Debt: 375B KZT EBITDA: 2,393B KZT	0.92x (Failed to deduct cash reserves from total debt)	Subtracted 1,818B KZT cash & bank deposits from 2,193B KZT gross debt. Verified true Net Debt = 375B KZT.	0.16x (Verified)
Net Profit Margin (%)	KMG 2025 Income Statement Net Profit: 1,072B KZT Revenue: 9,371B KZT	18.5% (Confused EBITDA margin with net margin)	Corrected formula to Net Profit / Consolidated Revenue (1,072 / 9,371). Confirmed Net Profit Margin.	11.44% (Verified)
EBITDA Margin (%)	KMG 2025 Financial Report EBITDA: 2,393B KZT Revenue: 9,371B KZT	25.5% (Accurate arithmetic)	Cross-checked against PwC audit report Note 6. Confirmed operational cash flow strength.	25.54% (Verified)
State Tax Contribution	KMG 2025 Annual Report Tax & Payments Page 45	1.5 Trillion KZT (Outdated 2023 figure)	Audited against 2025 tax receipts. Updated to reflect true national budget contribution of 2.3T KZT.	2.3 Trillion KZT
ABAI System Scale	KMG Digitalization Brief Section 'Tech Upgrade' p. 178	All upstream wells in Kazakhstan	Corrected AI overgeneralization. Verified exact waterflooding injection cell count across OMG/MMG/KOA assets.	1,660 Cells

Section 2: Week 9 Career Capability Portfolio Matrix

Mapping human, business, digital, and meta-capabilities against KMG's POLC Operating Model.

Capability Domain	Core Skills & Descriptors	Operating Model Link (Week 4 Task)	KMG Evidence & Context	Strategic Value Add
Human Capabilities	Empathy, Ethical Judgment, Labor Negotiations	Task 3: Mangystau Workforce Restructuring (Human Only)	Managing 52,000+ workers; fulfilling sovereign social mandate in Western KZ.	High: AI cannot hold legal accountability or negotiate labor stability.
Business Capabilities	Financial Audit, Strategic Transfer, Compliance	Task 1: CAPEX Allocation (Human + AI) & Transfer Analysis	Subsidizing domestic crude supply while cross-subsidizing via export netbacks.	High: Prevents catastrophic copying of Shell's price skimming strategy.
Digital Capabilities	AI Model Verification, Telemetry Audit, Prompting	Task 7: ANPZ Predictive Maintenance (AI Executes + Review)	Auditing ABAI machine learning alerts on 1,660 waterflooding cells.	High: Prevents false-positive shutdowns of critical refinery units.

Meta-Capabilities	Adaptability, Transfer Stress-Testing, Fallback Design	Task 8: Reservoir Pressure Optimization & Fallback Protocols	Designing manual operator logbook fallback for SCADA server failures.	High: Guarantees operational resilience during cyber or network outages.

Section 3: Consolidated Evidence Folder (IEEE Referencing)

[1] JSC NC 'KazMunayGas', *Annual Report 2025: Global Standards. National Priorities*, Astana, Kazakhstan, Apr. 28, 2026.

[2] Shell plc, *Annual Report and Accounts 2025: Deliver More Value with Less Emissions*, London, UK, Mar. 11, 2026.

[3] JSC NC 'KazMunayGas', *Consolidated Financial Statements for the Year Ended December 31, 2025*, PwC Audit Report, Astana, Mar. 5, 2026.

[4] Ministry of Energy of the Republic of Kazakhstan, *Order No. 171 on State-Regulated Petroleum Price Caps*, Official Gazette, Astana, 2026.

[5] KMG Engineering, *ABAI System Operational Brief and Digitalization Roadmap*, Astana, Kazakhstan, 2026.

[6] KAZENERGY Association, *National Energy Report 2023*, Astana, Kazakhstan, 2023.

[7] G. Yeshenkulova, N. Ibadildin, and M. Tleubayeva, *Business Administration*, Almaty: ZIALY BASPASY, 2026.

[8] Business Administration Course Text, *Introduction to Business*, 2nd ed. Houston, TX, USA: Rice University, 2024.